

A Multihead Attention Self-Supervised Representation Model for Industrial Sensors Anomaly Detection

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Abstract—Industrial sensors capture critical information for intelligent manufacturing maintenance. To promote equipment upgrading and manufacturing processes, intelligent decisions, and information learning play an important role. Although deep learning methods historically obtain excellent results, there is always a tradeoff between fine-tuning existing networks or designing models from scratch for sensor data processing. In this article, we propose the multihead attention self-supervised (MAS) representation model, which is a self-supervised learning-based sensor feature extraction network. To the best of our knowledge, this is the first time a self-supervised contrastive learning method using positive samples that represent multidimensional industry sensor data is being used for anomaly detection. We review alternative data augmentation methods proposed for better-representing sensor sequence data. We use this insight to design a new structure that adapts to the temporal characteristics of the application. We apply our method to a real-world water circulation system that uses a variety of industrial sensors. The effectiveness of the proposed MAS methods is demonstrated.

Index Terms—Anomaly detection, multihead attention self-supervised (MAS) feature representation method, sensor time-series data, the industrial Internet of things (IIoT).

I. INTRODUCTION

INDUSTRIAL Internet of things (IIoT) has been rapidly developing in the last few years, e.g., in smart factories, unmanned logistics sorting, and collaborative production. IIoT integrates industrial production equipment, industrial sensors, and monitoring devices. Intelligent analysis is embedded into different industrial production parts through wired and wireless transmission data. The processing and analysis of data are particularly important in industrial production processes [1]. And these important data are captured by industrial sensors. The collected information extracted from various datasets provides valuable information for subsequent evaluation and decision-making. An implementation of IIoT is the timely detection of system abnormalities via sensor data, which must be meaningful to ensure the safety and stability of industrial production.

Data generated by many sensors are widely transmitted. Fig. 1 depicts an industrial scenario of industrial production processes. There are a variety of different sensors that can collect industrial data, such as temperature and vibration. These data are dynamic and high-dimensional, processed and stored on edge computing via data transfer, and further transferred to the cloud by network technology [2]. Some tasks, such as status monitoring and anomaly detection, have been further implemented. Data processing methods are often adapted to mine information, but effective analysis is very important for anomaly detection or traffic analysis of an industrial system.

Data acquired by industrial sensors are usually in the form of a time series. The status information is represented via a single sensor, such as current data, temperature data, vibration data, and other data from various types of sensors. In some cases, the cooperation of multisensor information is required. For example, it may be difficult for a single sensor to capture a system-level abnormality, but the abnormality can be analyzed from data collected by multiple sensors. For a single sensor, the data currently obtained often have a particular correlation with data collected previously, indicating that the data are time-dependent.

Data analysis is important for IIoT status surveillance. Anomaly detection is one of the key implementations. Abnormal

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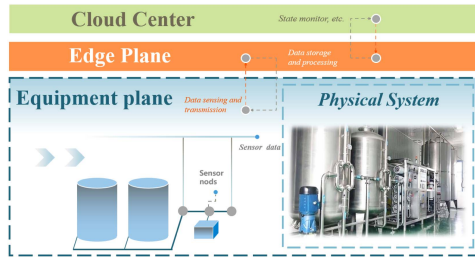


Fig. 1. Framework diagram of the IIoT. In industrial sites, data are collected by sensors and transmitted to the edge plane. And further, it is transmitted to the cloud plane through the network to accomplish some tasks, such as state monitoring. Therefore, effective data processing and analysis will aid in the mining of sensor data features and the information contained within them, which is of great importance for industrial production.

data include values that are too large or too small and can be defined by using expert knowledge. However, due to complex environments and operating conditions, abnormalities are often related to normal values. Hardware anomalies, such as various equipment failures, and anomalies caused by environmental disturbances are common in industrial systems. These industrial anomalies can disrupt the entire industrial system's production, and may even result in extremely significant mishaps. As a result, abnormalities in the system should be captured timely and accurately by using many sensors. This is also an important task of industrial internet intelligence and security.

In a nutshell, there are two types of anomaly detection approaches: traditional machine learning methods and deep learning methods. Traditional anomaly detection methods include time-series models, such as autoregressive moving average models [3], Markov models [4], and Kalman filters [5]. These methods do not generally have high detection accuracy and are also not good enough for high-dimensional long-sequence time-series feature extraction. In deep learning, neural networks apply to anomaly detection and show better detection performance than traditional methods. In the training network phase of supervised learning, training data must be labeled. Label samples for industrial data require manual labeling by experienced engineers, which is costly and time-consuming. If the quantity of negative samples marked throughout the training process is insufficient, the unbalanced training results will induce inadequate learning. Unsupervised learning refers to solving the problem of pattern recognition on unlabeled datasets. In anomaly detection, prediction-based or reconstruction-based methods are the most used. Unsupervised learning can far exceed supervised learning in terms of training set size. It is normally important to employ a normal dataset during training, and the proportion of the abnormal data in training samples may cause distinct accuracies. As a result, in real-world scenarios, that means the dataset must be carefully screened and selected. For example, Malhotra et al. [6] introduced an LSTM-based encoder-decoder model that learns to represent feature vectors of normal time-series behavior for the sensor anomaly detection problem and then applies reconstruction error to detect abnormalities. These unsupervised methods rely on the reconstruction of the input from the output or prediction tasks. However, prediction or

reconstruction approaches may not be necessary for representation learning.

It is hard work to train the model when the training set has abnormal data. Therefore, we consider that contrast learning is appropriate when applied to the field of anomaly detection. Meanwhile, Yann LeCun also believes that self-supervision is an important point for future AI research [7]. He thinks that self-supervised learning (SSL) is one of the most promising ways to build background knowledge and approximate a form of common sense in AI systems. Background knowledge can be considered as data generality representations. In the pre-training stage, the SSL model captures general representations through certain specific tasks or ingenious design by using unlabeled data. It is difficult to label all the data for AI systems, but self-supervision pretraining can learn the representation of unlabeled data. So we start to do some preliminary work in the field of industrial anomaly detection using self-supervised representation learning methods.

Despite artificial intelligence technology in different fields sharing some common intuition and model structure, details for specific fields are quite different. Simply adopting methods from other fields usually leads to bad performance. Hence, in practice, algorithms designed for specific fields should be applied to make the model effective. Specifically, for sensor data, we need careful design to modify the algorithms including data preprocessing, data augmentation, representation, and network structure.

In this work, we propose the multihead attention self-supervised (MAS) representation model, a novel end-to-end approach for self-supervised representation-based anomaly detection method. The MAS method allows the existence of outliers in the training set, which is more in line with the characteristics of actual collected data. The method does not depend on the assumption of a statistical distribution of the data. The main contributions of our article are as follows.

- 1) To the best of our knowledge, this is the first time a self-supervised representation contrastive learning method with positive samples is used for anomaly detection of industrial sensors. The network extracts general features of time-series data via a set of tasks. After the feature extraction work is completed in the pretraining stage, the process is applied to the anomaly detection task.
- 2) For the enhancement of time-series data, we propose three different enhancement strategies: adding noise, random masking, and flipping data. These methods are compared via experiments. Since abnormal samples can be regarded as a way of data enhancement and the number of abnormal samples is typically modest in real-world scenarios, there is no need to remove abnormal data in the pretraining stage.
- 3) A sensor data feature extraction network is proposed. The network can extract effective and generic features from time series data. In industrial production environments, there are also many sensors with sample values scattered over a wide range. Only raw sensor data are required when utilizing the MAS model.

The rest of this article is organized as follows. We review related work in Section II and describe our approach in detail in

Section III. We describe the dataset and perform experimental analysis in Section IV, and summarize our work in Section V.

II. RELATED WORK

In this section, we review the existing approaches for industrial anomaly detection. Since we also use a contrastive learning approach applied for anomaly detection tasks, as contrastive learning is the key technology of SSL, we also introduce it to anomaly detection.

A. Industrial Data Analysis

Anomaly detection techniques for IIoT have received much attention in recent years. Anomaly detection methods in IIoT can be divided into traditional machine methods and deep learning methods. Rodrigo et al. [8] proposed autoregressive integrated moving average model (ARIMA) to detect anomalies in sensors. The ARIMA method can transform a dynamic time series into a stationary one, and then predict future values based on past values by setting prediction intervals as criteria for anomaly detection. However, this method is typically applied to a one-dimensional time series, while the accuracy deserves further improvement.

A growing number of machine learning approaches are being applied to anomaly detection problems with the development of deep learning. Some neural networks, such as long short-term memory (LSTM) network, have been extensively researched and applied to time series data. Wu et al. [2] proposed a LSTM-Gauss-Bayes method for outlier detection for IIOT. They combined LSTM and Gaussian Bayesian models to introduce model prediction errors into a normally distributed Naive Bayesian model to detect anomalies.

LSTM is relatively lightweight, however, it cannot be parallelized during training. The attention mechanism focuses on a part of the information while ignoring other parts of the information. This methodology has been successful in transformers [9]. The attention mechanism is effective and can be trained in parallel; it is being gradually applied to anomaly detection in time-series data. Liu et al. [10] proposed a LSTM model of convolutional neural networks (CNNs) based on attention mechanisms for federal learning frameworks, resulting in data acquisition of fine-grained features. Li et al. [11] proposed a model using multihead attention mechanism to detect anomalous behavior in ICS network traffic data when smart charging piles. The model does not require preprocessing of the data, it can extract high-dimensional information between the original flows better than many other methods (such as CNN-LeNET [12], CNN-sVGG [13], CNN-sAlexNET [14], SVM-rbf [15], random forest [16], and decision tree [17]).

In addition to the approaches discussed previously, there are methods for detecting data anomalies based on generative adversarial networks (GAN) [18] or variational autoencoders (VAE) [19]. Both GAN model and VAE model are generative models that use training data to build real data distributions, and then use the learned models and distributions to generate modeling to model new data. Gong et al. [20] used a memory module to improve the autoencoder, where the most relevant memories are

recovered for reconstruction. The model makes no assumptions about the data type and in unusual instances, can increase the reconstruction error. Zhou et al. [21] proposed a variational LSTM network based on reconstructed representations for the industrial big data network intrusion problem. The work learns low-dimensional feature representations from high-dimensional raw data to identify not only the presence of anomalies, but also the class of anomalies. In addition, Qi et al. [22] addressed the issue of cyber-attacks in industrial logistics networks and proposes the multiaspect data streams detection method. This method combines the techniques of locality-sensitive hashing, isolation forest, and PCA. On the dataset UNSW-NB15, it achieves state-of-the-art performance. Su et al. [23] combined a Gaussian mixture model with a one-dimensional CNN to detect instance anomalies for servers in large data centers. It has also been validated and achieves the best performance on the ByteDance server dataset. Lin et al. [24] proposed an anomaly detection model that combines VAE and LSTM. The VAE model is used to construct robust local features over a brief window, and the LSTM method is used to estimate long-term correlations over VAE-inferred features. Although GAN model and VAE model sometimes have high precision for anomaly detection, the computational cost is high and such reconstruction may not be necessary for feature learning. Instead, various network structures are used to enhance data feature learning. For the problem of anomaly detection in industrial sensors, it is very important to design a reasonable model for representation learning.

B. Self-Supervised Representation Learning

Self-supervised representation learning has been rapidly developing in recent years and has been applied to the fields of image processing [25], video processing [26], and natural language processing [27] with good results.

Contrastive learning is a technique that allows a model to learn which data points are similar or different to understand the general characteristics of a dataset. In general, self-supervised representation learning is realized by constructing positive samples and negative samples and measuring the distance between positive and negative samples. The frequent loss function is InfoNCE [28], which seeks to bring the distance between positive sample pairs closer and the distance between negative sample pairs farther.

Many current contrastive learning approaches concentrate on vision processing and natural language processing. In the field of computer vision, methods of self-supervised representation learning are gradually gaining attention from researchers. Dorsch et al. [29] suggested using jigsaw puzzles to build an auxiliary task. By segmenting the image and predicting the relative positions of these parts, the network is trained. Methods based on contrast learning have produced positive results in recent years. Chen et al. [30] began by emphasizing the significance of various data enhancement methods before arguing that introducing non-linear transformations between data representation and contrast loss can improve the quality of the learned representations. Recently, generative-based self-supervised representation learning methods have gradually received attention from researchers.

TABLE I
TABLE OF SYMBOLS, THE CORRESPONDING MEANINGS OF
DIFFERENT SYMBOLS

Symbols	Meaning
$(x_1, \dots, x_n)$	Data collected by different sensors
$(x_i, \dots, x_{\text{seq}})$	Slicing of data $(x_1, \dots, x_n)$
v, v'	Data after different augmentation processes
f_θ, f_δ	Feature extractor
y, y'	Representation extracted by the feature extractor
g_θ, g_δ	Projector
p_θ	Predictor
q, z	The output of the predictor and the projector
τ	Delay rate
γ	Noise rate

He et al. [31] proposed a mask-based autoencoder to pretrain the backbone network via an asymmetric encoder-decoder structure; the literature [32] improved on this basis by predicting the features of the masked region rather than reconstructing the original image.

At present, there are few kinds of research on contrast learning in the field of time series anomaly detection. In literature [33], Deldari et al. first proposed the method of contrastive learning for time-series change point detection. In this research, a temporal convolutional network [34] is used to extract sequence features. The effect of batch size, code size, and windows size on the outcomes are investigated.

Finding appropriate auxiliary activities is critical for boosting model training accuracy in self-supervised representation learning. In the pretraining stage, various auxiliary tasks in the video, image, and natural language processing are designed to increase the ability of model feature extraction. However, applying contrastive learning methods to industrial time-series data instead poses some unique challenges. We attempt to address the problem of using self-supervised representation learning to provide effective representations for anomaly detection.

III. METHODOLOGY

We introduce a self-supervised representation contrastive learning model [35].

From the previous analysis, we know that industrial sensor data is time-dependent. When an abnormality occurs in the system, many sensors may gather abnormal data concurrently. To build more accurate and efficient detection models, we need to be able to capture time-dependent characteristics and use information collected from multiple sensors. At the same time, a self-supervised representation training strategy is adopted to learn more representation of the data.

In this work, we use anomaly detection task to test the performance of the network. The model shows good performance in the anomaly detection task. The following section will describe the details. The symbols' definitions are shown in Table I.

A. Data Representation Learning

A feature extractor, which maps time-series features into a compact embedding representation, is at the foundation of contrastive learning. The structure of the feature extractor will

be described later. The overall structure of contrastive learning is shown in Fig. 2.

The data collected by industrial sensors is a continuous sequence $(x_1, \dots, x_n)$, with each moment being a vector of multisensor. A vector sequence $(x_i, \dots, x_{\text{seq}})$ sampled from $(x_1, \dots, x_n)$, and the network generates two different enhanced data $v \triangleq t(x)$ and $v' \triangleq t'(x)$ by using various data augmentation methods. The data augmentation approaches will be introduced later. Subsequently, the data representations y and y' are obtained after the feature extraction network f_θ and f_δ , respectively. The feature q_θ is obtained by y via the projector and the predictor, and the feature z_δ is obtained by y' via the projector. Finally, we calculate the ℓ_2 loss between the feature q_θ and the feature z_δ

$$\mathcal{L}_{\theta, \delta} \triangleq \|\bar{q}_\theta - \bar{z}_\delta\|_2^2 \quad (1)$$

where, $\bar{q}_\theta \triangleq q_\theta / \|q_\theta\|_2$ and $\bar{z}_\delta \triangleq z_\delta / \|z_\delta\|_2$. The loss function in (1) is also the loss function in Fig. 2. The significance of using the ℓ_2 loss function is that representations of the same property are closer. In addition, the weight update method of f_θ is stochastic gradient descent, and the weight update method of f_δ is as follows:

$$\delta \leftarrow \tau \delta + (1 - \tau) \theta \quad (2)$$

where, τ is a decay rate, its ranges from 0 to 1. After completing the training, we obtain the sequence feature extraction network f_θ . For further information on the specific process, see Algorithm 1.

B. Data Augmentation

We use data augmentation for contrastive learning to compare positive data, e.g., by constructing negative sample pairs as in [33]. This article proposes three data enhancement approaches for industrial sensor data processing. The following are introduced.

1) *Adding Noise*: The mean and standard deviation of the time-series data gathered by each sensor are calculated separately. The approach for introducing Gaussian white noise is as follows:

$$x' = x + \gamma \cdot \text{noise}(0, \text{std}) \quad (3)$$

where, x' is the enhanced data, $\gamma \in [0, 1]$ is noise rate, $\text{noise}(0, \text{std})$ represents white Gaussian noise with a mean of 0 and a standard deviation of std.

2) *Data Flip*: For the sliced sensor data $(x_i, \dots, x_{\text{seq}})$, the data at each moment is a vector. There are two methods for flipping data. One method flips the series in temporal order and gets the flipped sequence $(x_{\text{seq}}, \dots, x_i)$. An alternate approach is to transpose by dimension, and the resulting data sequence is $(x'_i, \dots, x'_{\text{seq}})$, where x'_i, x'_{seq} is the transpose of x_i, x_{seq} .

3) *Data Mask*: Mask or replace a portion of the data in the time series at random. First, use the Bernoulli distribution to determine the segment to be masked in the data. Each training epoch has a different mask position to improve the network's ability to extract characteristics from the data.

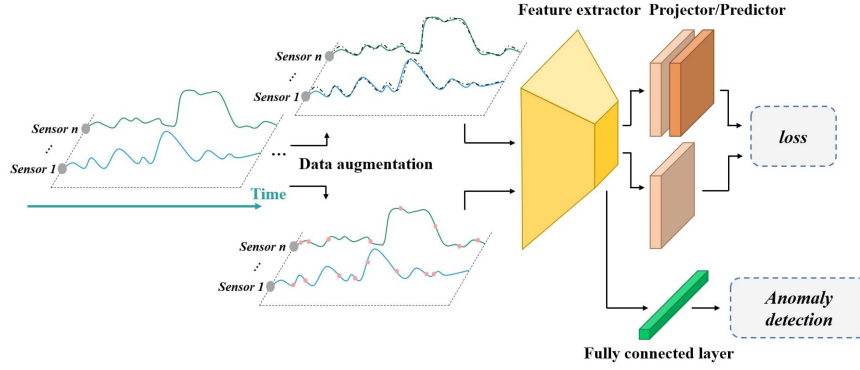


Fig. 2. Overall architecture of the MAS model. The data collected from the sensors are first enhanced via means of data augmentation, followed by a feature extractor to obtain representations of the data. The output features are obtained from these representations by running them through a projector/predictor head. The features are normalized before the l_2 loss is calculated. The significance of using the l_2 loss function is to make the features obtained by the predictor closer to the features of the projector head. The network's projector or predictor head is made up of two linear layers, a BN layer, and an activation function. After the feature extractor finishes training, a fully connected layer is added to complete the data anomaly detection.

Algorithm 1. Multi-Head Attention Self Supervised Feature Representation Model.

Input: Time-series data from sensors $X = \{x_n\}_{n=1}^N$

Output: A feature extractor f_θ

Slice the time-series data into sequences $(x_i, \dots, x_{seq})$

Get the sequence set $S = \{s_i\}_{i=1}^m$

while Training **do**

 Sample a batch from sequence set: $\mathcal{B}$

for $x \in \mathcal{B}$ **do**

 1. Perform two various data augmentation for sequence x and obtain enhanced data v and v' :
 $v \triangleq t(x)$, $v' \triangleq t'(x)$.

 2. The enhanced data v is sent through the feature extractor f_θ , projector g_θ and predictor p_θ to obtain the features q

 3. The enhanced data v' is sent through the feature extractor f_δ , projector g_δ to obtain the feature z

 4. Calculate the l_2 loss between q and z

 5. Update neural network parameters

end

end

while Fine-Tuning **do**

 1. Fixing the weight of f_θ , adding fully connected layer

 2. Training the weights of fully connected layers on real datasets

 3. Testing the accuracy of anomaly detection in the test set

end

C. Feature Extraction Network

1) *Dilated Casual Convolution*: Industrial sensor data are time-series data, and the determination of anomalies often requires analysis from a period before the anomaly. Neural networks require larger receptive fields to establish longer temporal

associations. The use of CNN leads to a proliferation of layers and model parameters, while recurrent neural network or LSTM has sequential inputs and has gradient explosion or slow training problems.

We employ dilated casual convolution [34] instead of general convolution and obtain a large perceptual field while using fewer network layers and model parameters in the MAS model. Causal convolution allows the output of the net at moment t to be only associated with the input before the moment t , as shown in Fig. 4. This operation is achieved by zero padding. This can avoid the influence of future information on the network output at the current moment.

2) *Multihead Attention*: The attention mechanism process can be described as calculating scores by queries and keys, and then multiplying the scores with values to obtain the output. The queries, keys, values, and output are all vectors.

The features extracted by dilated causal convolution are inputted into the multihead attention mechanism. The input then undergoes a linear mapping that yields the values of queries (Q), keys (K), and values (V). The dimension of Q and K is d_k [31], and the dimension of the values is d_v . After calculating the dot product of the queries and keys, we can obtain the weight of values through the softmax layer

$$\text{attention} = \text{softmax} \left(\frac{Q \cdot K^T}{\sqrt{d_k}} \right) V \quad (4)$$

where d_k is the scaling factor to prevent the result of the dot product from being too large.

The multihead attention mechanism is an improvement over the attention mechanism, which establishes different projection information in different spaces. The input matrices are projected differently and stitched together after many output matrices are obtained

$$\begin{aligned} \text{MultiHead}(Q, K, V) &= ZW^O \\ Z &= \text{Concat}(\text{head}_1, \dots, \text{head}_h) \end{aligned} \quad (5)$$

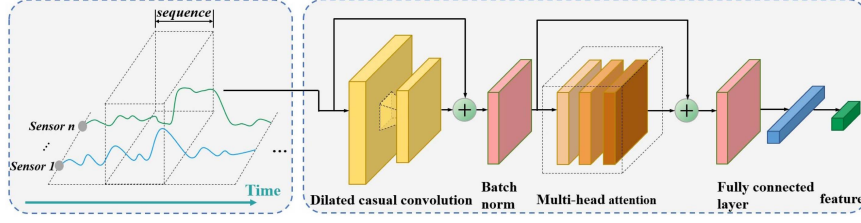


Fig. 3. Structure of the feature extractor. The data collected from industrial sensors are first sliced and sent through a dilated casual convolution with residual connection, then through a batch norm layer. The extracted features are processed via a multihead attention network with residual connections, followed by a fully connected layer, to obtain a representation of the data.

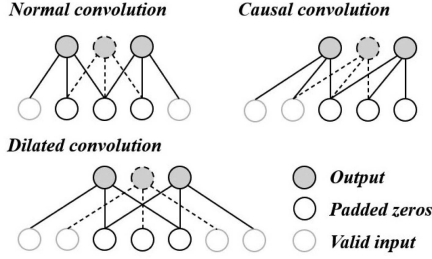


Fig. 4. Difference between normal convolution, causal convolution, and dilated convolution. Convolution with a kernel size of 3 is shown. The output of each layer of the causal convolution is obtained from the corresponding unknown input of the previous layer and the input of its previous positions. Dilated convolution skips part of the input and the filter can be applied to regions larger than the length of the filter itself so that a large perceptual field can be obtained.

where, $\text{head}_i = \text{attention}(QW_i^Q, KW_i^K, VW_i^V)$, $Q \in R^{n \times d_k}$, $K \in R^{n \times d_k}$, $V \in R^{n \times d_v}$. A linear layer, behind the multihead attention mechanism, is used as the output of the entire network. For features extracted by dilated casual convolution, the multihead attention model is used to capture the implied features between the multidimensional data.

3) **Network Structure:** The feature extractor's overall structure is shown in Fig. 3, which includes dilated casual convolution, batch norm model, RELU, and multihead attention mechanism. For the data collected from industrial sensors, the data are first sliced into sequences, and then the features are extracted through dilated casual convolution. The dilated casual convolution has a large perceptual field and a time-dependent feature that can process sensor input more effectively. The convolved features are then fed into the multihead attention mechanism. The multihead attention model aids in the extraction of characteristics from multidimensional data. The batch norm and residual modules serve to increase training stability in the self-supervised representation learning process. The end of the network structure is a fully connected layer for feature mapping. As a result, the data gathered from the industrial sensors are extracted as characterization information during network processing.

IV. EXPERIMENTS

In this section, experiments are designed to evaluate the MAS method to demonstrate the effectiveness of our proposed method.

TABLE II
SENSOR DETAILS, INCLUDING SENSOR TYPE, DATA ACQUISITION UNIT, AND QUANTITIES OF DIFFERENT SENSORS

Sensor type	Data unit	Number
Vibration acceleration	g	2
Current	Ampere	1
Pressure	Bar	1
Temperature	The degree Celsius	1
Thermocouple	The degree Celsius	1
Voltage	Volt	1
RateRMS	Liter per minute	1

A. Data Description

To evaluate the performance of our method in an industrial sensor environment, a publicly available dataset SKAB [36] is used. The dataset contains eight sensor data collected from real water circulation systems. The types of sensors and data collection units are shown in Table II.

The dataset adopted in our work has various types of sensors, including acceleration, current, and others. The physical quantities and ranges of data collected by sensors are different, and data are time-dependent. The information provided by these sensors reflects the overall operational status of the experimental platform system. The sensors are commonly used in industry; we admit that the scale of the system is small compared to the real industrial environment. However, it can reflect the characteristics of industrial sensor networks. In addition, we apply to preprocess the data before training and testing. The specific steps are: calculating the mean and standard deviation values of each sensor data and then performing normalization.

B. Parameter and Structure

We experiment with the MAS model and some other algorithms on the same dataset. In our experiments, we use Pytorch and Nvidia 2080Ti for model acceleration. We use the SGD optimizer during model training. The target decay rate is $\tau = 0.999$.

In the feature extractor, the output dimension of the two-layer dilated casual convolution is 32. Subsequently, the output and input dimensions are kept the same size after a multihead attention mechanism is applied. Then going through a batch norm and a linear layer, we get a ten-dimensional output. Projection and prediction both have the same structure, the dimension of the projection input is ten and the dimension of the output is eight. The prediction input is eight and the dimension of output is eight.

We compare other machine learning methods with our proposed MAS method, including LSTM, multivariate state estimation technique (MSET), multiscale convolutional recurrent encoder-decoder (MSCRED), LSTM variational autoencoder (LSTM-VAE), Hotelling's T-squared statistic (T-squared), autoencoder, and isolation forest.

C. Evaluation Metrics

We apply accuracy, precision, recall, and F1 to determine the performance of the model comprehensively [2]. The detailed formulas for the metrics are as follows:

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{FN} + \text{FP} + \text{TN}} \quad (6)$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (7)$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \quad (8)$$

$$F_1 = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad (9)$$

where TP, TN, FP, and FN are four classification results, denoting true positive, true negative, false positive, and false negative, respectively.

The recall metric represents how many abnormal samples are found. Precision represents how many of the samples judged to be abnormal are true. F_1 represents the combined accuracy of model detection, which is a combination of recall and precision.

D. Results and Analysis

1) Data Augmentation: We address the problem of anomaly detection by contrastive learning. Anomaly detection aims to distinguish the difference between normal and abnormal sample features. In general, there is an imbalance of samples, and the amount of abnormal events is small. Contrastive learning can get a good feature extraction with few negative samples. Data augmentation is an important method in contrastive learning. During the training phase, contrastive learning reduces the distance between similar samples and increases the distance between negative samples, to realize the representations of normal samples and abnormal samples. The model is adopted to extract features of sensor data. Through the method of data augmentation and contrastive learning, the model will extract better features in the pretraining phase.

We explore how different data augmentation strategies affect feature extraction. In this article, we apply three data enhancement techniques: adding noise, randomly performing mask, and data flipping. The data flip includes an up and down flip (UD flip) and a left and right flip (LR flip). The simulation results are shown in Table III.

For using two data enhancement methods, the combination of noise and flip is better than the combination of mask and flip, but both are slightly weaker than the comparative learning approach of noise and mask in terms of the F1-score. UD flip outperforms LR flip in terms of results. This indicates that it is easier for the network to extract features when only changing the temporal order of the sequence data without adjusting the spatial

TABLE III
EFFECT OF DIFFERENT DATA AUGMENTATION METHODS ON THE RESULTS. THERE ARE THREE MAIN DATA AUGMENTATION METHODS: ADDING NOISE, RANDOM MASK, AND DATA FLIPPING

Noise	ACC	Precision	Recall	F1-score
noise+mask	0.7258	0.8119	0.7808	0.7961
noise+flip_ud	0.6918	0.7665	0.7907	0.7784
noise+flip_lr	0.3967	0.6588	0.2467	0.3589
mask+flip_ud	0.5386	0.7586	0.4786	0.5869
mask+flip_lr	0.3619	0.5732	0.2670	0.3643
noise+(mask,flip_ud)	0.6920	0.8046	0.7267	0.7637
noise+(mask,flip_lr)	0.6465	0.6885	0.8834	0.7739
mask+(noise,flip_ud)	0.6093	0.7504	0.6435	0.6929
mask+(noise,flip_lr)	0.4826	0.7153	0.4062	0.5182

features. As can be observed from simulation results, there is a certain improvement in detection accuracy when employing multiple data enhancement with LR flip compared with only using the two data augmentation methods with LR flip. For the multiple data enhancement methods using UD flip, there is no significant improvement in the highest accuracy compared with the corresponding two data augmentation approaches using UD flip, and both are the same in F1-score. We can see that using three means of data augmentation is not always as effective as using two. We consider that as the quantity of data augmentation methods grows, modified segments of original data will also increase. Too much change in the data will damage its properties. The model can effectively learn representations if augmented data is within a certain range. When too much data augmentation is used, the effectiveness of model extraction representation will be reduced.

Contrastive learning aims to learn representations that are invariant under different distortions. Intuitively, if anomaly samples appear in the training phase, their representations will also be learned as well as those of the normal. While contrastive approaches of SSL learn representations by minimizing the distance between two augmented views of the same data point (positive pairs) and maximizing views from different data points (negative pairs), recent noncontrastive SSL shows remarkable performance without negative pairs. Compared to contrastive methods, these noncontrastive methods do not require large batch sizes or memory queues [25] to provide negative pairs. Therefore, excluding the negative pairs, the noncontrastive learning that we use is generally more efficient and conceptually simple.

We use the method of adding noise and random masking as an example to explain the effectiveness of data augmentation. After adding noise to the data, data will fluctuate within a range. The random mask method selects a set of data points and masks them. These masked data points may be normal or abnormal. When the network converges to a nonzero nontrivial solution, the entire network can work. The network can learn representations of both anomaly and normal features from the training data. This is also verified by the relevant experiments described in the article. The network can achieve a relatively high F1-score with noise and masks, demonstrating the effectiveness of the design strategy.

2) Effect of Augmentation Parameters: We investigate the effect of different enhancement parameters on the simulation results. Analyze the effect on the imitation results when different

TABLE IV

EFFECT OF DIFFERENT MASK RATE PARAMETERS ON THE RESULTS. THE KEY POINTS INCLUDING ACC, PRECISION, RECALL, AND F1-SCORE ARE REPORTED

Mask rate	ACC	Precision	Recall	F1-score
0.01	0.2999	0.4651	0.1492	0.2261
0.05	0.7274	0.8367	0.7479	0.7898
0.1	0.72025	0.8313	0.7558	0.7872
0.2	0.7258	0.8119	0.7808	0.7961
0.3	0.6766	0.7161	0.8745	0.7874
0.4	0.7166	0.8131	0.7611	0.7862
0.5	0.3235	0.5132	0.2356	0.3333
0.6	0.5084	0.7094	0.4779	0.5711
0.7	0.3232	0.5122	0.2469	0.3332

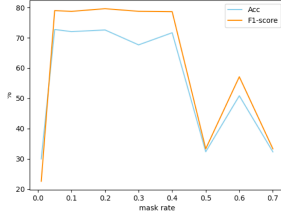


Fig. 5. Effect of the mask rate parameter on accuracy and F1-score. It can be seen that the effects of this parameter on Acc and F1-score are basically synchronized. The Acc and F1-score of anomaly recognition are low when the mask rate is small or large. The mask rate between 0.05 and 0.4 is better.

TABLE V

EFFECT OF DIFFERENT NOISE RATE PARAMETERS ON THE RESULTS. THE KEY POINTS INCLUDING ACC, PRECISION, RECALL, AND F1-SCORE ARE REPORTED

Noise rate	ACC	Precision	Recall	F1-score
0.01	0.269	0.3877	0.1163	0.1789
0.05	0.7258	0.8115	0.7804	0.7961
0.1	0.7126	0.8142	0.7517	0.7817
0.2	0.5644	0.7218	0.5921	0.6557
0.3	0.6230	0.7663	0.6468	0.7015
0.4	0.3618	0.5520	0.3608	0.4364
0.5	0.5550	0.7290	0.5574	0.6317
0.6	0.3789	0.5693	0.3818	0.4571
0.7	0.3981	0.5937	0.3830	0.4656

values of the mask are given separately. For example, mask = 0.2 means that 20% of the data are masked out. The evaluation results are shown in Table IV and Fig. 5. As the mask values range from 0.01 to 0.7, the MAS model works better when the mask ratio is between 0.05 and 0.4. The maximum F1-score is obtained when the mask ratio is 0.2. When the mask proportion is too low, the model has a hard time learning data features, which results in lower anomaly detection accuracy. It is also difficult for the network to extract useful information because a large amount of information is masked away when the mask proportion is higher.

Similarly, the impact of noise magnitude on the results is examined further down. In the experiments, the value of the mask rate is given as 0.2 and the noise ratio is given in the range of 0.01–0.7. Table V and Fig. 6 illustrate the evaluation results. The simulation results demonstrate that when the noise rate is small and the F1-score of anomaly detection is low, the network cannot successfully extract features from the data through contrast learning. It can be inferred that the network does not learn valuable features from data. When the noise rate level is between 0.05 and 0.1, relatively high detection accuracy can

TABLE VI

RESULTS OF ABLATION EXPERIMENTS

Method	Acc	Precision	Recall	F1-score
Remove dilated casual CNN	0.3301	0.5180	0.3105	0.3883
Remove multihead attention	0.3313	0.5193	0.3155	0.3925
Remove BN layer	0.6449	0.7178	0.7212	0.7195

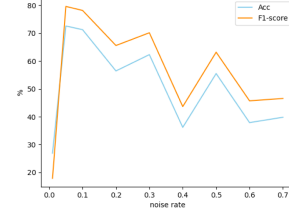


Fig. 6. Effect of noise rate parameter on accuracy and F1-score. Acc and F1-score have basically the same trend. The Acc and F1-score of anomaly recognition are low when the noise rate is small or large. When the noise is between 0.05 and 0.3, the network can achieve a better result.

be obtained. Because noise reduces the proportion of relevant information in the sensor network, the useful features extracted by the network will decrease as the noise rate grows, resulting in a fall in overall anomaly detection accuracy.

3) Ablation Study: Experiments for the ablation study are shown the performance of the modules in Table VI.

The ablation study results show that the major indexes have obviously reduced when removing dilated casual CNN module or multihead attention module. The addition of the BN layer is helpful to improve the metric of the model. Experimental results show the effectiveness of the modules.

4) Algorithm Complexity Analysis: The algorithm's complexity is examined below. In industry, the ability to anomaly detection quickly is critical. We analyze the proposed algorithm's time complexity. We refer to article [9] for the complexity analysis of the algorithm. The algorithm can simply be thought of as consisting of two parts: the dilated causal convolution module and the multiheaded attention module. The higher proportion of time complexity is the multiheaded attention module. The complexity of the multiheaded attention mechanism is $O(nd_{mh}^2 + n^2d_{mh})$. As a comparison, the time complexity of LSTM is $O(nd_l^2)$. When $d_l = d_{mh}$, the complexity of our proposed model is slightly higher than that of LSTM. In practice, because of its better effect on feature extraction, MAS can be experimentally verified to have a lower dimension of d . In addition, unlike LSTM, the dilated causal convolution and multihead attention mechanisms can perform operations in parallel in practice. Furthermore, to determine the speed of detection of the sensor sequence by the algorithm, we run our program on the CPU (i7-10875 h). For a given time sequence, the single anomaly detection time is 0.00134 s. The LSTM ($d_l = d_{mh}$) runtime is 0.00086 s, Bi-LSTM runtime is 0.00168, LSTM ($d_l > d_{mh}$) is 0.00249 s, the running time of LSTM-VAE is 0.00471 s. Experimental comparisons with these classical algorithms prove that our algorithms can be effectively deployed in industrial scenarios.

5) Overall Results: We compare several alternative baseline approaches on the dataset. The F1-score is an important metric

TABLE VII

COMPARISON OF OUR PROPOSED MAS MODEL WITH OTHER ALGORITHMS IN SOME METRICS, INCLUDING ACC, RECALL, PRECISION, AND F1-SCORE

Algorithm	Acc	Precision	Recall	F1-score
MSET [37]	0.7944	0.6772	0.7991	0.7332
LSTM [38]	0.7658	0.6959	0.6047	0.6471
MSCRED [39]	0.7824	0.7069	0.6899	0.6982
LSTM-VAE [40]	0.7488	0.7283	0.4615	0.5651
T-squared [41]	0.7357	0.6812	0.4744	0.5593
Autoencoder [42]	0.7160	0.7083	0.3342	0.4542
Isolation forest [43]	0.7008	0.6898	0.2791	0.3974
MAS(ours)	0.7258	0.8119	0.7808	0.7961

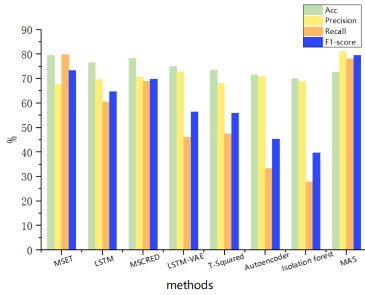


Fig. 7. Comparison of the MAS method with other methods in several metrics of accuracy, precision, recall, and F1-score. It can be seen that our method has the highest score on the F1-score. This indicates that our model trainable strategy can extract the features well from the industrial sensor data.

for expressing anomaly detection performance because it represents the harmonic mean of precision and recall. F1-score is only high when both precision and recall are high. Table VII and Fig. 7 show and compare the evaluation results. Both isolation forests and LSTM are unsupervised learning methods. LSTM can produce good results, outperforming isolation forests. The MAS method obtains the highest F1-score among all methods, with 81.19% for precision, 78.08% for recall, and 79.6% for F1-score. Overall, the findings show that our method can effectively extract the temporal properties of sensor data and can better distinguish abnormal values from normal ones.

E. Discussion

Although self-supervised representation learning does not provide data labels in the pretraining stage, it merely allows the network to learn to extract features of the data on its own through comparison learning. When applied to anomaly detection tasks, a high F1-score can be obtained with simple training, compared to other methods.

V. CONCLUSION

In this article, we focus on detecting system anomalies from sensor data in real-world scenarios. In our work, we first present a contrastive learning strategy for training feature extractors, which can be applied to detect anomalies in industrial sensor data. Subsequently, we propose three different data augmentation approaches, so that outliers in the training sequence do not have to be used. Extensive experiments prove that our MAS model is effective. Finally, we present an end-to-end neural network to handle the problem of extracting potential time-dependent relationships in high-dimensional

data. Compared with other methods, our proposed MAS approach achieves good results.

In the future, we will further investigate the application of self-supervised representation learning in anomaly detection. In addition, we are looking into other approaches for detecting anomalies in industrial sensors. At the same time, we will further develop our model to achieve better performance and be more adaptable in a wide range of usage scenarios.

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